Supporting Information

Kinney and Atwal 10.1073/pnas.1309933111

SI Text

Theorem 1. *Let X and η be (not necessarily independent) random variables and define $Y = X + \eta$. For any invertible function h, the random variable $\xi = Y - h(X)$ obeys the Markov chain condition $X \leftrightarrow h(X) \leftrightarrow \xi$.*

Proof: This follows trivially from the fact that, because h is invertible, $p(\xi \mid h(X), X) = p(\xi \mid h(X))$, which is what it means for $X \leftrightarrow h(X) \leftrightarrow \xi$ to be a Markov chain. Note that the definition of ξ is actually irrelevant here. □

Theorem 2. *Every self-equitable dependence measure $D[X;Y]$ is invariant under all invertible transformations of X and Y.*

Proof: If $X \leftrightarrow f(X) \leftrightarrow Y$ is a Markov chain and the functions h_1 and h_2 are invertible, then

$$X \leftrightarrow h_1(X) \leftrightarrow X \leftrightarrow f(X) \leftrightarrow Y \leftrightarrow h_2(Y) \leftrightarrow Y \qquad \textbf{[S1]}$$

is also a Markov chain. Extracting various Markov subchains and invoking Eq. **3** of the main text, we see that for every self-equitable dependence measure D,

$$X \leftrightarrow h_1(X) \leftrightarrow Y \Rightarrow D[X;Y] = D[h_1(X);Y], \qquad \textbf{[S2]}$$

$$Y \leftrightarrow h_2(Y) \leftrightarrow h_1(X) \Rightarrow D[Y;h_1(X)] = D[h_2(Y);h_1(X)]. \qquad \textbf{[S3]}$$

Invoking the symmetry of D we get $D[X;Y] = D[h_1(X);h_2(Y)]$, which is what we aimed to prove. □

Theorem 3. *Every DPI-satisfying dependence measure $D[X;Y]$ is self-equitable.*

Proof: If $X \leftrightarrow f(X) \leftrightarrow Y$ is a Markov chain, so is

$$f(X) \leftrightarrow X \leftrightarrow f(X) \leftrightarrow Y. \qquad \textbf{[S4]}$$

Extracting Markov subchains and using Eq. **4** of the main text, we see than for any DPI-satisfying measure $D[X;Y]$,

$$X \leftrightarrow f(X) \leftrightarrow Y \Rightarrow D[X;Y] \leq D[f(X);Y] \qquad \textbf{[S5]}$$

$$f(X) \leftrightarrow X \leftrightarrow Y \Rightarrow D[f(X);Y] \leq D[X;Y] \qquad \textbf{[S6]}$$

and so $D[X;Y] = D[f(X);Y]$, proving that D is self-equitable. □

Theorem 4. *Every dependence measure $I_F[X;Y]$ that can be written as*

$$I_F[X;Y] \equiv \int dx\, dy\, p(x)p(y)F\left(\frac{p(x,y)}{p(x)p(y)}\right), \qquad \textbf{[S7]}$$

where F is a convex function on the nonnegative real numbers, satisfies DPI. Note that such measures of dependence are called "F-information" measures (1).

Proof: Let $X \leftrightarrow Y \leftrightarrow Z$ be a Markov chain, so that $p(x,z) = \int dy\, p(x \mid y)p(y,z)$ and $p(x) = \int dy\, p(x \mid y)p(y)$. Then

$$I_F[X;Z] = \int dx\, dz\, p(x)p(z)F\left(\frac{p(x,z)}{p(x)p(z)}\right) \qquad \textbf{[S8]}$$

$$= \int dx\, dy\, dz\, p(x \mid y)p(y)p(z)F\left(\frac{\int dy\, p(x \mid y)p(y,z)}{\int dy\, p(x \mid y)p(y)p(z)}\right) \qquad \textbf{[S9]}$$

$$\leq \int dx\, dy\, dz\, p(x \mid y)p(y)p(z)F\left(\frac{p(y,z)}{p(y)p(z)}\right) \qquad \textbf{[S10]}$$

$$= \int dy\, dz\, p(y)p(z)F\left(\frac{p(y,z)}{p(y)p(z)}\right) \qquad \textbf{[S11]}$$

$$= I_F[Y;Z], \qquad \textbf{[S12]}$$

where in Eq. **S10** we have used Jensen's inequality (2). This proves that $I_F[X;Z] \leq I_F[Y;Z]$ and thus that I_F satisfies DPI. □

Noise in the X Variable. The supplemental material of ref. 3 as well as follow-up work (4) has suggested that R^2-equitability should actually be extended to include cases where noise is present in both the X and Y variables. This is formalized by replacing Eq. **1** with

$$D[X + \zeta; Y] = g\left(R^2[f(X);Y]\right), \qquad \textbf{[S13]}$$

where ζ is an additional noise term. It is clear, however, that this more general definition does not permit nontrivial solutions. This stems from the fact that the left side of Eq. **S13** depends on ζ whereas the right side does not. Any measure $D[X + \zeta; Y]$ would therefore have to be invariant to the amount of noise ζ in the X coordinate. For example, consider $f(X) = X$, $\eta = 0$ and assume the noise ζ is very large and does not depend on X. Eq. **S13** would require $D[X;X] = D[X + \zeta; X] \approx D[\zeta; X]$, implying the value assigned by D to the identity relationship must be the same as the value assigned to a relationship in which the two variables in question are independent.

MIC and I_{MIC} Are Usually Identical. As mentioned in the main text, we were unable to extract I_{MIC} values from the MIC estimator of Reshef et al. (3) because the source code has not, as of this writing, been made publicly available. However, we were able to extract I_{MIC}, n_X, and n_Y values from the MIC estimator of Albanese et al. (5).

For over 99% of the MIC computations shown Fig. 3 *A* and *B* of the main text, the MIC algorithm chose binning schemes having only two bins on either the x or the y axis. As mentioned above, $MIC\{x;y\} = I_{\mathrm{MIC}}\{x;y\}$ in such cases. It therefore appears that the primary tests of equitability reported by Reshef et al. (3) provide virtually no information about how the normalization procedure in the definition of MIC affects MIC's equitability.

The common occurrence of $MIC\{x;y\} = I_{\mathrm{MIC}}\{x;y\}$ is also reflected in Fig. S5. On the linear, parabolic, and sinusoidal relationships we found $Z_{\mathrm{MIC}} = 1$ (corresponding to $\min(n_X, n_Y) = 2$) for every one of the 37,500 relationships tested. Thus, I_{MIC} and MIC exhibited identical power profiles. However, on the circular and checkerboard relationships we consistently observed $Z_{\mathrm{MIC}} > 1$ (i.e., $\min(n_X, n_Y) > 2$) when noise was sufficiently low; Z_{MIC} dropped back down to 1 as noise increased.

Tellingly, I_{MIC} actually outperformed MIC in tests of statistical power. It appears that right at the noise level at which highly structured relationships become difficult to discern, the binning scheme chosen by MIC starts to fluctuate. The normalization factor Z_{MIC} therefore fluctuates, causing MIC values to become

noisier—and thus less powerful—than the underlying mutual information estimate I_{MIC}. This suggests that the normalization procedure in Eq. **7** of the main text, which transforms I_{MIC} into MIC, may actually be harmful for quantifying dependencies.

Computational Methods. The KNN mutual information estimator of Kraskov et al. (6) was downloaded from http://bsp.teithe.gr/members/downloads/Milca.html. Except for k, which we varied depending on context, the KNN estimator was run with default settings.

Except where stated otherwise, MIC estimates were computed with the software described by Reshef et al. (3), downloaded from www.exploredata.net, using default parameters (including $\alpha = 0.6$). All other MIC values were computed using a version of the estimator of Albanese et al. (5) (with parameters $\alpha = 0.6$, $c = 15$) that was edited to also report the values of I_{MIC}, n_X, and n_Y.

All other analysis was performed in Matlab. Our Matlab scripts, along with our edited version of the Albanese et al. MIC estimator, are available at https://sourceforge.net/projects/equitability/.

1. Csiszár I, Shields PC (2004) *Information Theory and Statistics: A Tutorial* (Now Publishers, Hanover, MA).
2. Cover TM, Thomas JA (1991) *Elements of Information Theory* (Wiley, New York).
3. Reshef DN, et al. (2011) Detecting novel associations in large data sets. *Science* 334(6062):1518–1524.
4. Reshef DN, Reshef Y, Mitzenmacher M, Sabeti P (2013) Equitability analysis of the maximal information coefficient, with comparisons. arXiv:1301.6314v1 [cs.LG].
5. Albanese D, et al. (2013) Minerva and minepy: A C engine for the MINE suite and its R, Python and MATLAB wrappers. *Bioinformatics* 29(3):407–408.
6. Kraskov A, Stögbauer H, Grassberger P (2004) Estimating mutual information. *Phys Rev E Stat Nonlin Soft Matter Phys* 69(6 Pt 2):066138.

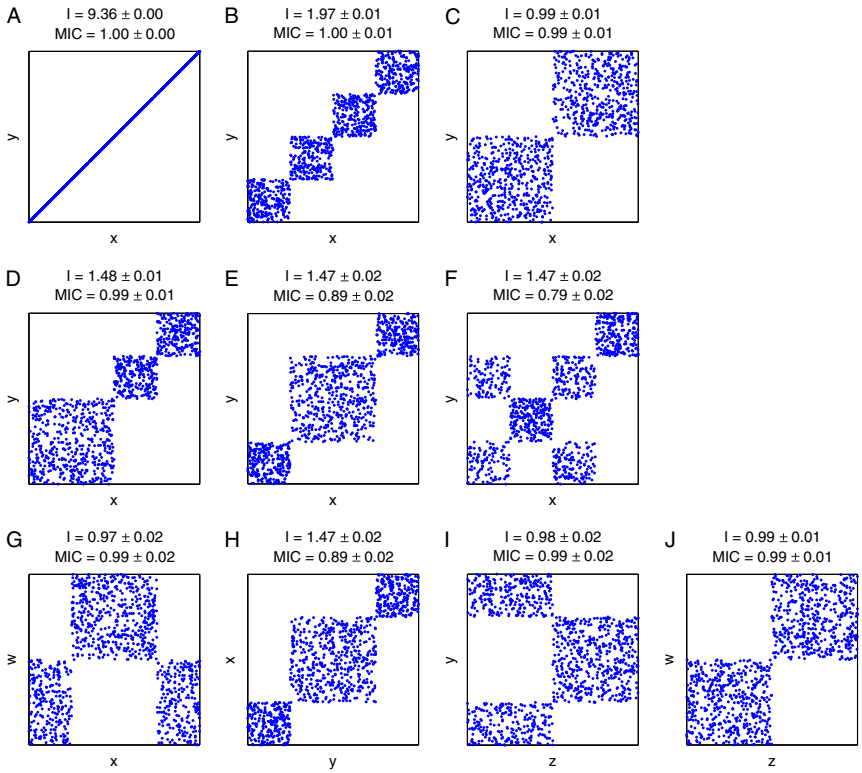

Fig. S1. (*A–J*) Estimates of mutual information and MIC computed for $N = 1{,}000$ data points sampled from the joint probability distributions shown in Fig. 2. Mutual information estimates were computed using the KNN estimator of Kraskov et al. (6) with $k = 1$. MIC estimates were computed using the estimator of Reshef et al. (3) with default parameters. Values for both mutual information and MIC are reported as mean ± SD over 100 replicates.

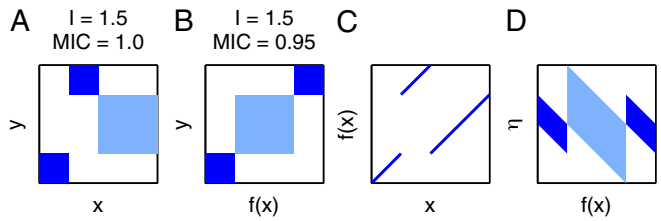

Fig. S2. MIC violates self-equitability. The relationships in *A* and *B* result from $Y = f(X) + \eta$ with function f and noise term η respectively defined in *C* and *D*. Note that the noise term η, as required, depends on $f(X)$ but not otherwise on X. MIC is seen to violate self-equitability (Eq. **3**) because $MIC[X;Y] \neq MIC[f(X);Y]$.

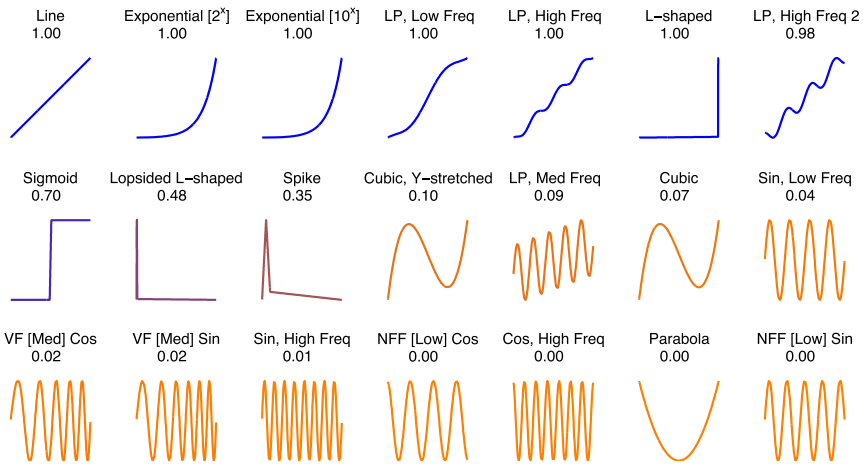

Fig. S3. Functions f used in the simulations described in Fig. 3 and Fig. S4 and listed in Table S1. Functions are colored according to each one's (indicated) monotonicity (Fig. 3*E* and Fig. S4*H*).

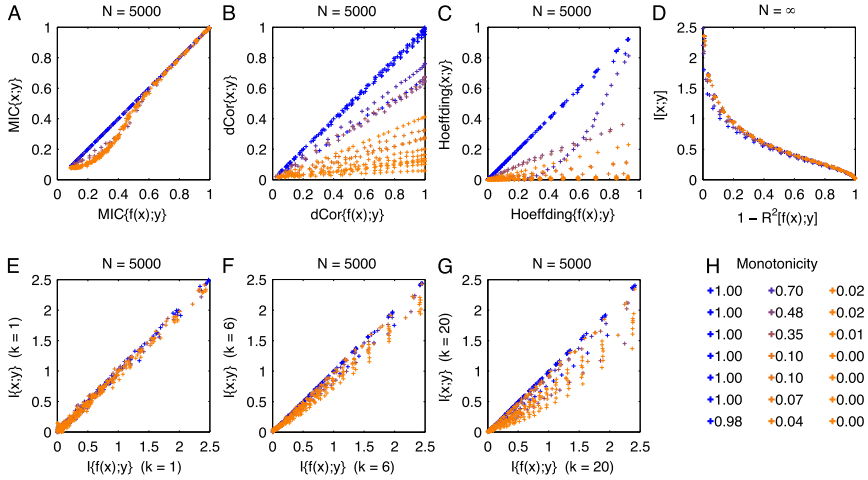

Fig. S4. Additional equitability tests of MIC, dCor, Hoeffding's D, and the KNN mutual information estimator. Noisy relationships were simulated as in Fig. 2 *B* and *D*; see Table S1 for details. (*A*) The MIC estimator of Reshef et al. (1) violates self-equitability on the data simulated for Fig. 2*B* of the main text: Specifically, $MIC\{x;y\} < MIC\{f(x);y\}$ for sufficiently nonmonotonic functions f. This downward bias of MIC is particularly pronounced for $MIC\{f(x);y\}$ values between 0.2 and 0.4. (*B* and *C*) A downward bias for nonmonotonic functions, indicative of non–self-equitability, is also seen in similar tests of (*B*) dCor (2) and (*C*) Hoeffding's D (3). (*D*) Extrapolation of Fig. 2*D* of the main text to datasets of infinite size. Here, mutual information values were computed semianalytically using $I[X;Y] = H[Y] - H[\eta]$ where H is entropy (4). (*E*–*G*) Mutual information estimates $I\{x;y\}$ returned by the KNN algorithm using (*E*) $k = 1$, (*F*) $k = 6$, and (*G*) $k = 20$. Increasing values of k result in decreasing adherence to self-equitability. This is most apparent for nonmonotonic functions, which are more readily blurred by smoothing than are monotonic functions. (*H*) Correspondence between color and monotonicity; same as Fig. 3*E*.

1. Reshef DN, et al. (2011) Detecting novel associations in large data sets. *Science* 334(6062):1518–1524.
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3. Hoeffding W (1948) A non-parametric test of independence. *Ann Math Stat* 19(4):546–557.
4. Cover TM, Thomas JA (1991) *Elements of Information Theory* (Wiley, New York).

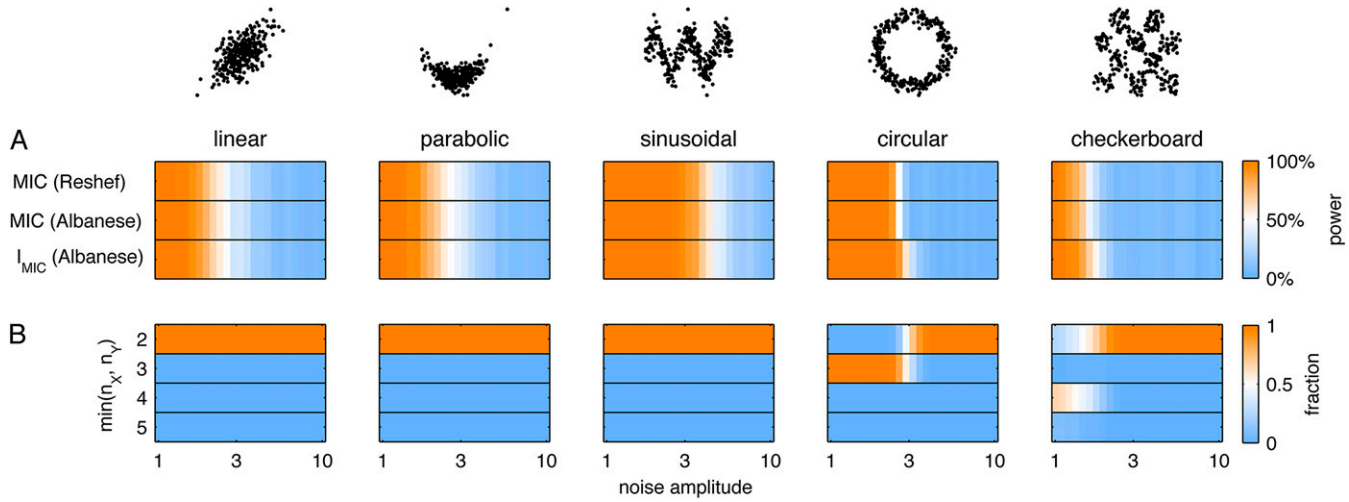

Fig. S5. Power comparisons for I_{MIC} and two MIC estimators. (*A*) Power, estimated as in Fig. 4, for the MIC estimator of Reshef et al. (3), for the MIC estimator of Albanese et al. (5), and for the corresponding I_{MIC} values computed by the Albanese et al. estimator. (*B*) Each heat map shows the fraction of simulations, at each noise level, for which the Albanese et al. MIC estimator used an $n_X \times n_Y$ binning scheme with the indicated value of $\min(n_X, n_Y)$. This value determines the normalization constant $Z_{MIC} = \log_2(\min(n_X, n_Y))$.

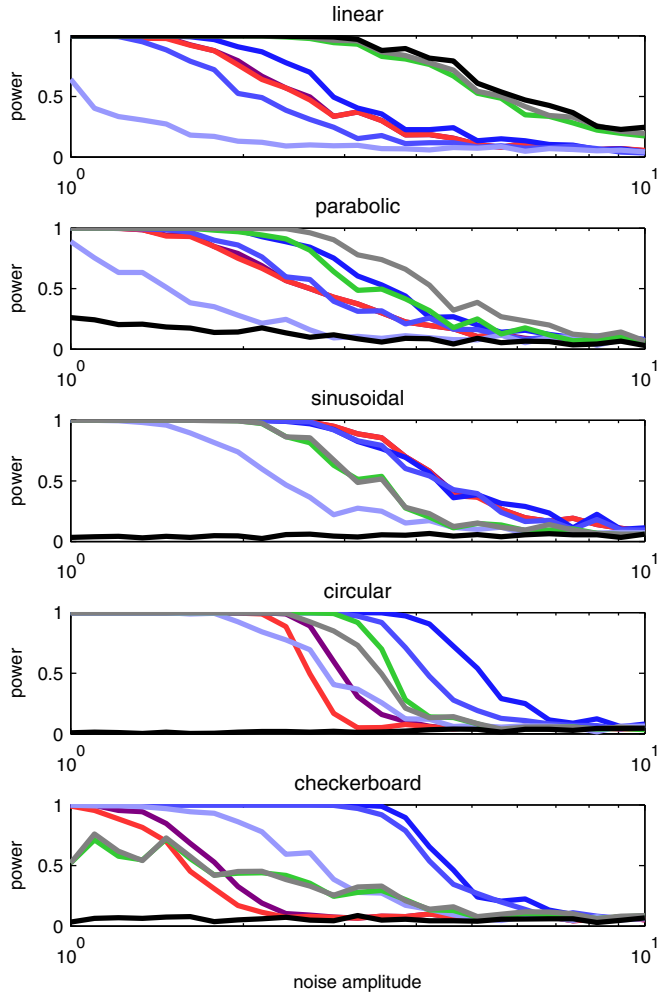

Fig. S6. Power curves corresponding to the heat maps shown in Fig. 4 and Fig. S5. For each of the relationships tested (linear, parabolic, sinusoidal, circular, and checkerboard), power vs. noise is plotted for eight different statistics: R^2 (black); dCor (gray); Hoeffding's D (green); MIC computed using the algorithm of Reshef et al. (3) (red); I_{MIC} computed using the modified algorithm of Albanese et al. (5) (purple); and mutual information computed using the KNN estimator with $k = 1$ (light blue), $k = 6$ (medium blue), or $k = 20$ (dark blue).

Table S1. Noisy functional relationships simulated for Fig. 3 and Fig. S4, ordered according to monotonicity as described in the Fig. 3 legend

Description	Monotonicity	N	X	$f(X)$
Line	1.00	1,000	[0, 1]	X
Exponential [2^X]	1.00	1,000	[0, 10]	2^X
Exponential [10^X]	1.00	1,000	[0, 10]	10^X
LP, low frequency	1.00	1,000	[0, 1]	$\frac{1}{5}\sin(4(2X-1))+\frac{11}{10}(2X-1)$
LP, high frequency	1.00	1,000	[0, 1]	$\frac{1}{10}\sin(10.6(2X-1))+\frac{11}{10}(2X-1)$
L shaped	1.00	1,000	[0, 1]	$\begin{cases} X/99 & \text{if } X \le \frac{99}{100} \\ 99X-98 & \text{if } X > \frac{99}{100} \end{cases}$
LP, high frequency 2	0.98	1,000	[0, 1]	$\frac{1}{5}\sin(10.6(2X-1))+\frac{11}{10}(2X-1)$
Sigmoid	0.70	1,000	[0, 1]	$\begin{cases} 0 & \text{if } X < \frac{49}{100} \\ 50\left(X-\frac{1}{2}\right)+\frac{1}{2} & \text{if } \frac{49}{100} \le X < \frac{51}{100} \\ 1 & \text{if } X \ge \frac{51}{100} \end{cases}$
Lopsided L shaped	0.48	500*	[0, 1]	$\begin{cases} 200X & \text{if } X < \frac{1}{200} \\ -198X+\frac{199}{100} & \text{if } \frac{1}{200} \le X < \frac{1}{100} \\ -\frac{X}{99}+\frac{1}{99} & \text{if } X \ge \frac{1}{100} \end{cases}$
Spike	0.35	1,000	[0, 1]	$\begin{cases} 20X & \text{if } X < \frac{1}{20} \\ -18X+\frac{19}{10} & \text{if } \frac{1}{20} \le X < \frac{1}{10} \\ -\frac{X}{9}+\frac{1}{9} & \text{if } X \ge \frac{1}{10} \end{cases}$
Cubic, Y stretched	0.10	1,000	[−1.3, 1.1]	$41(4X^3+X^2-4X)$
LP, medium frequency	0.09	500*	[0, 1]	$\sin(10\pi X)+X$
Cubic	0.07	1,000	[−1.3, 1.1]	$4X^3+X^2-4X$
Sin, low frequency	0.04	250	[0, 1]	$\sin(8\pi X)$
VF [med] cos	0.02	1,000	[0, 1]	$\cos(5\pi X(1+X))$
VF [med] sin	0.02	500*	[0, 1]	$\cos(6\pi X(1+X))$
Sin, high frequency	0.01	1,000	[0, 1]	$\sin(16\pi X)$
NFF [low] cos	0.00	250*	[0, 1]	$\cos(7\pi X)$
Cos, high frequency	0.00	500*	[0, 1]	$\cos(14\pi X)$
Parabola	0.00	1,000	$\left[-\frac{1}{2}, \frac{1}{2}\right]$	$4X^2$
NFF [low] sin	0.00	1,000	[0, 1]	$\sin(9\pi X)$

These relationships correspond to the relationships simulated by Reshef et al. (3) for Fig. 2 *B–F*, as described in table S3 of ref. 3. Here we have abbreviated function names as follows: LP, linear + periodic; VF, varying frequency; and NFF, non-Fourier frequency. For each relationship listed, data $\{x_i, y_i\}_{i=1}^N$ were simulated, where $Y = f(X)+\eta$, X ranged over the indicated domain, f was one of the 21 functions shown, and η was uniform noise of amplitude equal to the range of $f(X)$ times one of these 24 relative amplitudes: 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.15, 1.3, 1.45, 1.60, 1.8, 2.1, 2.5, 3.1, 4.5, 6, 8, 10, or 20. We note that, as in the simulations Reshef et al. (3) used for their figure 2, X values were not distributed uniformly within the allowed range but rather were chosen so that the points $(X, f(X))$ were spaced evenly along the length of each curve in the x,y plane. This, for instance, is the reason the monotonicities of the "Cubic" and "Cubic, Y stretched" relationships are slightly different.

*Reshef et al. (3) also simulated these relationships using $N = 1{,}000$ data points; we performed simulations only with the number of data points indicated.

Table S2. The *X*,*Y* relationships simulated for the power calculations of Fig. 4 and Figs. S5 and S6

	Random variable	
Relationship type	X	Y
Linear	ξ	$\frac{2}{3}X + a\eta$
Parabolic	ξ	$\frac{2}{3}X^2 + a\eta$
Sinusoidal	$\frac{5}{2}\theta$	$2\cos(X) + a\eta$
Circular	$10\cos(\theta) + a\xi$	$10\sin(\theta) + a\eta$
Checkerboard	$10X_0 + a\xi$	$10Y_0 + a\eta$

Five hundred trial datasets were generated for each of these relationships at each of twenty-five different noise amplitudes distributed logarithmically between 1 and 10. For each dataset, statistics were computed on the "true" data $\{x_i, y_i\}_{i=1}^{N}$ as well as on "null" data, for which the indexes i on the y values had been randomly permuted. The power of each statistic was defined as the fraction of true datasets yielding a statistic value greater than 95% of the values yielded by the corresponding null datasets. ξ and η are random numbers drawn from the normal distribution $\mathcal{N}(0,1)$. θ is a random number drawn uniformly from the interval $[-\pi,\pi)$. (X_0, Y_0) is a pair of random numbers drawn uniformly from the solid squares of a 4×5 checkerboard, where each square has sides of length 1. a is the "noise amplitude" shown in Fig. 4 and Figs. S5 and S6.